Rohan Bahl

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EDUCATION

National University of Singapore Jan 2026

Bachelor of Computing (Hons), Computer Science (Specialisations: Artificial Intelligence & Database Systems)

- Special Programme: **NUS Overseas Colleges Toronto–Waterloo AY 24/25 Sem 2 Intake**

EXPERIENCE

Cortex AI Jan 2026 – Sep 2026

Founding Engineer Singapore

- Diagnosed and resolved a client-blocking pose jitter issue within 3 hours, unblocking 1.5 months of stalled R&D, by tracing the regression to compounding smoothing on velocity derivatives and re-stabilizing forward-kinematics pose reconstruction with raw-derivative 1-Euro filtering, inter-MCP distance constraints, and Jakobsen relaxation
- Replaced 3 weeks of manual hierarchical planning with a sub-minute automated workflow for annotation-task allocation across a 4-level Environment → Scene → Task → Operator hierarchy, by designing a greedy diversity-maximizing optimizer with profile-aware caps and all-pairs allocation, exposed through a FastAPI two-step claim preview/rollback workflow and a Catalogue-integrated planner UI
- Built a cross-platform mobile data capture app in under a week, enabling the company's end-to-end field data collection pipeline, by integrating native Swift/ARKit and Kotlin/CameraX/ARCore modules for per-frame LiDAR/depth capture behind a unified Dart interface, with background-queued multi-artifact resumable GCS uploads and Clerk-authenticated session contracts
- Designed and proposed a VLM-based temporal alignment algorithm for 3 unsynchronized cameras as an alternative to OCR/template-matching pipelines, cutting session sync time by 90% (20 min to under 2 min), by adapting binary search to a noisy Gemini oracle with consensus voting, transition-onset detection, and a scipy cross-correlation fallback
- Architected the clock-authority layer for a production multi-human Vive Tracker + ZED Box capture stack on Jetson Orin, eliminating multi-source drift across ZED frames, Vive tracker poses, and IMU samples for 50+ operators / 1000+ hours of data, by building an on-box FastAPI service that owns the single timebase, a PySide6/QML operator console with shared-memory preview from the QSG render thread, and cloud-backed trim and annotated-render pipelines feeding scenario-keyed GCS folders
- Eliminated daily upload failures for low-connectivity operators across SG, India, and the Philippines, saving 2-5 hours/month of manual bucket and DB cleanup across 3-4K file batches, by engineering browser-side GCS resumable chunked uploads with retry, duplicate detection, and offset recovery

The Tudor Group May 2025 – Aug 2025

Software Engineer Intern Singapore

- Architected and shipped a horizontally scalable, real-time alerting platform (4 Spring Boot microservices, ActiveMQ, Redis, WebSocket) that replaced the legacy Nimsoft system for the Site Reliability team, delivering sub-second UI updates with message deduplication and aggregation across high-volume Grafana alert streams
- Reduced Grafana infrastructure management time by 99% (10 hrs → 5 mins) by building an Ansible + Python automation framework using the Grafana REST API for dashboard migration, alert generation, and multi-environment config management across dev, staging, and prod
- Cut trade log extraction time by 95% (60s → 3s) and eliminated legacy licensing costs by delivering a FIX log viewer (React, TypeScript, TanStack Table) within 2 weeks, integrated with Spring Boot APIs and adopted by the global TradeOps team of 50+ users

Upside Robotics Aug 2024 – Mar 2025

Software Engineer Intern Waterloo

- Architected and deployed a serverless event-driven telemetry pipeline (Lambda, S3, DynamoDB) via IaC (Serverless Framework), validating and processing 5M+ sensor records per run (including image compression/classification) into structured time-series data for real-time fleet monitoring across 5+ autonomous robots
- Reduced manual field diagnostics by 30% by building a real-time fleet operations dashboard (Next.js, TypeScript, Mapbox) featuring live telemetry visualization and geospatial tracking across 5+ autonomous robots, enabling engineers to identify and resolve issues remotely
- Improved field data reliability by 40% over intermittent LTE networks by designing and implementing resilient firmware logic (batching and offline caching) in collaboration with hardware engineers, eliminating critical data loss incidents across 5+ autonomous robot deployments in production environments

PROJECTS

StoryForge (Cursor Hackathon 2025) Oct 2025 – Nov 2025

ML Engineer / Team Lead Singapore

Built a 2D AI-powered game engine using Google Gemini AI and fal.ai for procedurally generated worlds with consistent narratives, visuals, and dynamic audio via ElevenLabs voice agents

- Finalist, Top 4% (15 / 400 projects)

GUI Murphy (TikTok TechJam 2025) Aug 2025 – Sep 2025

ML Engineer / Team Lead Singapore

Engineered a multi-stage UI identification and consistency validation pipeline using OmniParser, CLIP, and multimodal LLMs (MLLMs) to automatically detect design inconsistencies across mobile interfaces

- Finalist, Top 4% (12 / 300 projects)

Smart Storybook (ETHGlobal SF 2024) Oct 2024 – Oct 2024

Backend Engineer San Francisco

A Web3 platform enabling creators to monetize intellectual property through NFTs

- Built a FastAPI backend integrating Pinata's IPFS API for decentralized asset storage; Won 2nd place, Best AI Application

SKILLS

- **Programming Languages:** Python, Java, JavaScript/TypeScript, C
- **Databases:** MongoDB, MySQL, PostgreSQL, DynamoDB
- **Frameworks & Libraries:** Spring Boot, FastAPI, Flask, React, Next.js, Streamlit, Serverless Framework, PyTorch
- **DevOps & Tools:** GitHub Actions, Ansible, Redis, ActiveMQ, Grafana, Hadoop (HDFS & MapReduce), Spark, Flink
- **Cloud & Infrastructure:** GCP, AWS (EC2, S3, Lambda, Cognito, CloudWatch), Vercel